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Measuring air quality over large urban areas: development and application of an air pollution index at the urban area of Naples

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Abstract

A daily air pollution index (PI) has been developed and implemented at the urban area of Naples (Italy). Data gathered from nine monitoring stations during 2001–2002 have been analysed and a PI has been developed and applied. The index aims at measuring the status of air pollution with respect to its effect on human health. It can be seen as a modified version of air quality index of Environmental Protection Agency taking into consideration the limit values ruling in Europe. A procedure to evaluate the PI at each monitoring site and on the overall urban area is reported. Additive effects of air pollutants have also been considered and the PI re-evaluated.
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1. Introduction

The interest in the development of indexes for measuring the air pollution in urban areas is demonstrated by the number of publications on this topic. Indexes are named as: air quality indexes (AQI) or pollution indexes (PI). Air quality monitoring data are more copious as more air monitoring networks are installed and implemented in urban areas. Several pollutants are generally monitored by each station and hourly or daily average concentration data are calculated. Therefore, air monitoring networks provide a large mass of data. As a consequence, their interpretation by public authorities and their presentation to population becomes a hard task. Moreover, the diffusion of sources of information, such as the internet, enables the synthesis of the large mass of data coming from air monitoring networks into few and easy to understand indexes.

The main objective of air quality indexes is to measure the air quality with respect to its effects on the human health. Typically this is a daily quality index. A daily AQI has been proposed by Environmental Protection Agency (EPA) (1999). It is defined with respect to the five main common pollutants: carbon monoxide (CO), nitrogen dioxide (NO_2), ozone (O_3), particulate matter (PM_{10}) and sulphur dioxide (SO_2). Pollutants concentrations are converted into a numerical index (AQI) which assumes values in the range 0–500. The overall range is subdivided into six ranges to which six categories of air quality correspond (Table 1). The conversion of the concentration of each pollutant to the corresponding numeric value of the AQI is not a simple linear standardisation. In fact, for each pollutant the breakpoint concentrations, corresponding to each

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Table 1
Breakpoints for the AQI of EPA (volumes are standardised at a temperature of 293 K and a pressure of 101.3 kPa)

	AQI	PM_{10} (24 h) ($\mu g\,m^{-3}$)	NO_2 (1 h) ($\mu g\,m^{-3}$)	CO (8 h) ($mg\,m^{-3}$)	SO_2 (24 h) ($\mu g\,m^{-3}$)	O_3 (1 h)[b] ($\mu g\,m^{-3}$)	O_3 (8 h) ($\mu g\,m^{-3}$)
Hazardous	301–500	396–563	2374–3903	35.5–58.7	1608–2673	808–1206	[c]
Very unhealthy	201–300	331–395	1244–2373	18.0–35.4	810–1607	408–807	249–747
Unhealthy	151–200	238–330		14.5–17.9	597–809	328–407	209–248
Unhealthy for sensitive groups	101–150	144–237	[a]	11.0–14.4	384–596	250–327	169–208
Moderate	51–100	51–143		5.2–10.9	91–383		129–168
Good	0–50	0–50		0–5.1	0–90		0–128

[a] NO_2 has no short term NAAQS.
[b] Areas are generally required to report the AQI based on 8-hour ozone values. In some cases AQI based on 1-hour would be more precautionary. In these cases both AQI would be calculated and the maximum of the two values reported.
[c] When 8-hour O_3 concentrations exceed AQI, values must be calculated with 1-hour O_3 concentration.

category, are not on a linear scale. Breakpoint concentrations have been defined by EPA on the basis of National Ambient Air Quality Standards (NAAQS), and on the results of epidemiological studies of the effect of single pollutants on human health.

It must be observed that the values of breakpoint concentration derived from epidemiological researches are affected by some uncertainty and change frequently as more information on health effects become available. At the same time standards or guidelines for air quality introduced by public authorities become always more stringent. As a matter of fact, different breakpoint concentration values and air quality standards are reported in the literature (WHO, 2000; EC, 1999, 2000, 2002). Moreover, in defining breakpoint concentrations it would be significant to consider local conditions. As a matter of fact, different areas of the world are characterised by different climatic conditions influencing the effect of atmospheric pollutants on human health and also the response of population to air pollution could be different. Therefore, it is not realistic to assume the same air PI as valid all over the world.

Another problem is the evaluation of the AQI referred to a site and not to a single pollutant. In the atmosphere several pollutants are present simultaneously and the effects on human health due to the simultaneous presence of different pollutants in the atmosphere should be considered. In the evaluation of the AQI corresponding to a station (or site), where more than one pollutant is monitored, the EPA procedure (EPA, 1999) assumes the maximum AQI among those for the pollutants monitored. Actually, the category corresponding to the pollutants with the highest AQI would be assumed at least as that corresponding to that location.

Knowledge of the effects of a mixture of air pollutants on human health is at present limited. An attempt in considering this effect in the evaluation of pollution indexes was proposed by Khanna (2000) using a constant elasticity of substitution function but the absence of epidemiological data did not allow the assumption of the proper values for the parameters contained in the function. A revised EPA air quality index (RAQI) was proposed by Cheng et al. (2004) and the concentration of other pollutants, apart from that with maximum AQI, was considered by introducing an entropy function. A uniform indexing scale using well pre-established air quality standards and at the same time accounting for local conditions assessed via statistical analysis of data recorded at each monitoring station was proposed by Kassomenos et al. (1999) and implemented at the Athens metropolitan area. Air quality indexes with unlike goals are also reported in the literature. A daily quality index was proposed by Albergamo et al. (1996) to show exceeding limit values.

A PI with the same objective has been applied assuming limit values established by EC (Guzzi, 2004). Air pollution indexes have also been proposed for air pollution forecasts as reported by Cogliani (2001).

In this paper a PI has been proposed and implemented at the urban area of Naples. A modified version of the index has been developed assuming additive effects of pollutants. The procedure to evaluate the PI on the overall urban area is also reported.

2. Definition of the pollution index

Air quality in the city of Naples is monitored by a network of nine fixed stations measuring conventional pollutants: CO, NO_2, NMHC, PM_{10}, O_3 and SO_2 (Murena et al., 2000). The nine stations can be classified as: a background station (NA1), two stations tracking air quality in residential areas (NA2 and NA3), four high-traffic-area monitoring stations (NA4–NA7) and two photochemical pollution monitoring stations (NA8 and NA9).

An analysis of the air quality in the urban area of Naples was carried out by the author (Murena et al., 2000). The results show how the area is polluted especially by PM_{10}, NO_2, CO and O_3 (in the summer). A monitoring campaign carried out in a deep street canyon showed the occurrence of a high concentration level for benzene also (Murena and Vorraro, 2003).

In this paper concentration data measured by air quality monitoring networks in the urban area of Naples in the years 2001–2002 have been analysed. The application of AQI proposed by EPA (Table 1) gave the results reported in Table 2. It can be observed that making reference to the year 2001, if EPA AQI is assumed, air quality was "*good*" in more than 87.7% of the days for all pollutants except for PM_{10}. Occurrence of higher pollution categories (unhealthy for sensitive groups and others) are very rare. The number of times the limit values established by EC (Table 3) are exceeded is reported in Table 4 for each pollutant. In 2001, percentages of times limit values were exceeded as follows: 1.4% for NO_2, 49.2% for PM_{10} and 7.5% for O_3 (8 h average). The situation in 2002 was better but limit values were exceeded several times. By comparing the results reported in Table 4 with those obtained applying the EPA AQI (Table 2) the following observations are evident: (i) the application of EPA AQI to the

Table 2
Application of EPA AQI to data measured in Naples

Pollution category	2001						2002					
	CO 8 h	NO_2 1 h	SO_2 24 h	PM_{10} 24 h	O_3 1 h	O_3 8 h	CO 8 h	NO_2 1 h	SO_2 24 h	PM_{10} 24 h	O_3 1 h	O_3 8 h
Good	87.7	—	99.6	50.8	94.7	94.7	96.6	—	100	83.9	95.6	98.7
Moderate	12.3	—	0.4	48.0	4.9	4.9	3.4	—	—	16.1	4.2	1.3
Unhealthy for sensitive groups	—	—	—	1.2	—	—	—	—	—	—	0.3	—
Unhealthy	—	—	—	—	0.1	0.1	—	—	—	—	—	—
Very unhealthy	—	—	—	—	0.3	0.3	—	—	—	—	—	—
Hazardous	—	—	—	—	—	—	—	—	—	—	—	—

Table 3
Limit values for protection of human health established by EC directives (volumes are standardised at a temperature of 293 K and a pressure of 101.3 kPa)

Pollutant	Limit value	Averaging period	Maximum number of exceeds
PM_{10} ($\mu g\,m^{-3}$)	50	24 h	35 per calendar year
	40	Calendar year	from 2010
	20		
NO_2 ($\mu g\,m^{-3}$)	200	1 h	18 per calendar year
	40	Calendar year	
CO ($mg\,m^{-3}$)	10	8 h[a]	
SO_2 ($\mu g\,m^{-3}$)	350	1 h	24 per calendar year
	125	24 h	3 per calendar year
O_3 ($\mu g\,m^{-3}$)	120	8 h	25 per calendar year averaged over 3 years

[a]Mobile average.

Table 4
Urban area of Naples: exceeds of limit values established by EC

	2001					2002				
	CO 8 h	NO_2 1 h	SO_2 24 h	PM_{10} 24 h	O_3 8 h	CO 8 h	NO_2 1 h	SO_2 24 h	PM_{10} 24 h	O_3 8 h
Number of exceeds of LV	0	566	0	420	54	0	466	0	153	39
% of exceeds with respect to data collected	0	1.4	0	49.2	7.5	0	0.8	0	16.1	4.9

data of Naples gives for about 100% of the days a pollution category corresponding to "*good*" or "*moderate*" (generally more than 90% belonging to "*good*" category); (ii) the concentration levels corresponding to the highest category (hazardous) are never reached; (iii) the results in terms of EPA AQI and the number of times the limit values were exceeded as observed in the urban area of Naples in the same years seem to differ.

Considering these observations it may be noted that EPA AQI does not consider the short-time effect of NO_2 (Table 1). The data of this pollutant contributes to the AQI only in the case of the occurrence of concentrations $> 1244\,\mu g\,m^{-3}$ belonging to the "*very unhealthy*" or "*hazardous*" category. Moreover, AQI proposed by EPA seems to contain an excess of categories for measuring highest concentration levels and a deficiency for lower ones. Therefore, it is necessary to define a different AQI.

The reference scale for each pollutant of the AQI proposed here, will use at the same time: (i) limit or target values established by European Community and (ii) breakpoint concentrations to which an effect to human health corresponds. Moreover, since the PI is also a tool intended to give information to the general public, the number of categories must be limited and their definition clear. In defining the breakpoint concentrations of the PI the following concentration levels were considered for each pollutant:

- *Target or guide value*: A value to be attained wherever possible, below which it can be assumed there is no effect both on human health and on environment. When available it corresponds to the limit value for environment protection (EC, 1999; 2000, 2002).
- *Limit value*: The limit to be avoided in order to prevent or reduce harmful effects on human health. It corresponds to limit value for human health protection (EC, 1999; 2000, 2002).
- *Risk level for sensitive groups*: Concentration level above which effects on sensitive groups are reported in the literature (EPA, 1999; Gilli et al., WHO, 2000).
- *Risk level for population*: Concentration level above which effects for all the population, and/or more significant effects for sensitive groups, are reported in the literature (EPA, 1999; Gilli et al., WHO, 2000).

The first two levels are established by EC directives while the second two come from epidemiological studies on single pollutants. In correspondence to the above reported concentration level the following pollution categories have been defined:

- *Good quality*: The concentration of the pollutant is below the target or guide value established for the protection of the environment by EC.
- *Low pollution*: The concentration is below the limit value established by EC for human health protection. These values are in agreement with the guidelines proposed by the WHO for individual substances based on effects other than cancer or odour/annoyance (WHO, 2000).
- *Moderate pollution*: The pollutant concentration is above the limit value established by EC. Effect on population would not be present on the basis of present knowledge.
- *Unhealthy for sensitive groups*: The concentration of the pollutant can have effect on sensitive groups (children, asthmatics, etc.).
- *Unhealthy*: The concentration of the pollutant can have effect on all the population and/or greater effects on sensitive groups.

With respect to EPA AQI five categories are defined instead of six, and conditions corresponding to concentration levels lower than limit values are described by two categories instead of only one.

The reference scale of the AQI (in the following indicated as pollution index (PI) proposed here, with the corresponding pollution categories, is reported for each pollutant in Table 5. The reference scale can assume values in the range 0–100.

The evaluation of the PI at station s for pollutant p ($PI_{s,p}$) is carried out by a linear interpolation of the reference scale values reported in Table 5.

$$PI_{s,p} = \left[\frac{PI_{hi} - PI_{lo}}{BP_{hi} - BP_{lo}}(C_p - BP_{lo}) + PI_{lo}\right]_{s,p} \quad (1)$$

where $PI_{s,p}$ is the value of the PI at site s for pollutant p; C_p the daily reference concentration of pollutant p; BP_{hi} the lowest breakpoint of pollutant p that is greater than or equal to C_p; BP_{lo} the highest breakpoint of pollutant

p that is less than or equal to C_p; PI_{hi} the PI value corresponding to BP_{hi} of pollutant p and PI_{lo} the PI value corresponding to BP_{lo} of pollutant p.

Concentration C_p is a daily reference concentration. Since the PI is a daily PI, the reference concentration C_p is the maximum value of hourly (NO_2, SO_2, O_3) or 8-hour (CO and O_3) average concentration or simply 24 h average concentration (PM_{10}). If concentration higher than those reported in Table 5 in correspondence to PI equal to 100 are to be measured, the maximum value of the PI is assumed (PI = 100).

To make the communication of the air quality to the population easy the use of clouds as symbols corresponding to the categories of pollution defined is proposed in Table 5.

3. Application of the pollution index

The application of the PI in Table 5 to data collected during years 2001/02 gave the results reported in Table 6. As can be observed the occurrence of the lowest values of concentration is now better distributed in two categories ("*good quality*" and "*low pollution*") with respect to results obtained with EPA AQI. However, the occurrence of higher polluted categories "*Unhealthy for sensitive groups*" and "*Unhealthy*" still seem to be too rare considering that the urban area of Naples is characterised by a high vehicular traffic level and a rather old vehicular fleet.

It is important to note that breakpoint concentrations of EPA AQI and of PI proposed in this paper have been established for single pollutants. Generally pollutants, when mixed, can have additive, synergistic or antagonistic effects on human health. At present the knowledge of these interactions among air pollutants is rudimentary. To evaluate the AQI at a monitoring station, where more pollutants are measured, the EPA procedure assumes for the site the maximum AQI of the single pollutants. Following the same procedure the PI at a site (PI_s) would be evaluated using

$$PI_s = \max\left(PI_{s,p}\right). \tag{2}$$

In this way PIs of other pollutants do not contribute to the evaluation of the PI of that site. As a matter of fact since PI has been defined on the basis of effects of

Table 5
Breakpoints of the Pollution Index proposed (volumes are standardised at a temperature of 293 K and a pressure of 101.3 kPa)

Pollution category	Symbol	PI	PM_{10} 24 h ($\mu g\,m^{-3}$)	NO_2 1 h ($\mu g\,m^{-3}$)	CO 8 h ($mg\,m^{-3}$)	SO_2 24 h ($\mu g\,m^{-3}$)	O_3 1 h ($\mu g\,m^{-3}$)	O_3 8 h ($\mu g\,m^{-3}$)
Unhealthy		100	500	1900	30	1000	600	500
Unhealthy for sensitive groups		85	238	950	15.5	500	324	223
Moderate pollution		70	144	400	11.6	250	240	180
Low pollution		50	50	200	10	125	180	120
Good quality		25	20	40	4	20	—	65

Table 6
Application of PI, proposed in this paper, to data measured in Naples

Pollution category	2001					2002				
	CO 8 h	NO_2 1 h	SO_2 24 h	PM_{10} 24 h	O_3 8 h	CO 8 h	NO_2 1 h	SO_2 24 h	PM_{10} 24 h	O_3 8 h
Good quality	72.4	6.7	52.7	10.2	56.3	92.6	6.4	77.8	30.1	37.5
Low pollution	27.6	79.0	47.2	47.4	36.8	7.4	88.6	22.2	55.4	57.8
Moderate pollution	—	13.1	0.1	41.6	6.3	—	4.7	—	14.5	4.7
Unhealthy for sensitive groups	—	0.8	—	0.9	—	—	0.4	—	—	—
Unhealthy	—	—	—	—	0.6	—	—	—	—	—

Percentage of occurrence of pollution categories.

single pollutants, a PI_s value higher than that evaluated by Eq. (2) would be more representative of the actual situation if additive or synergistic effects are present and a lower value, if antagonistic effects are present. Since antagonistic effects are rarer than synergistic or additive ones the actual value of PI_s would be at least equal to that evaluated by Eq. (2); but generally it would be higher.

If additive effects of the different pollutants are assumed the PI can be evaluated using a procedure similar to that applied in the field of industrial hygiene for verifying threshold limit values in the case of a mixture of air pollutants (Crowl and Louvar, 1990). The sum of the ratios between the reference daily concentration of pollutant p (C_p) and the bottom breakpoint concentration ($BPb_{p,c}$) corresponding to each category c for any pollutant p (see Table 5) is calculated for all the pollutant categories:

$$\sum_{p=1}^{n} \frac{C_p}{BPb_{p,c}}. \tag{3}$$

Then PI_s is evaluated by

$$PI_s = PIb_c \sum_{p=1}^{n} \frac{C_p}{BP_{p,c}}, \tag{4}$$

where PIb_c is the bottom value of PI corresponding to the highest pollution category c for which the sum in Eq. (3) is $\geqslant 1$.

With this hypothesis air quality data of Naples have been re-elaborated and PI_s re-evaluated. The results are reported in Table 7. A significant difference with respect to the results reported in Table 6 where additive effects were not considered can be observed. As a matter of fact, assuming additive effects the highest pollution categories ("*Unhealthy for sensitive groups*" and "*Unhealthy*") were measured in more than 10% of days both in 2001 and 2002. These results show that if additive effects are present they can strongly influence the evaluation of the air quality. Actually additive effects can be assumed if pollutants belong to the same category and have similar effects on human health. This probably is not the case for all five main pollutants considered in the PI proposed. In this case, the results reported in Table 7 represent an overestimation of the actual PI. A better knowledge of mutual interactions of air pollutants in determining effects on human health would determine a more precise use of the procedure proposed in Eqs. (3) and (4).

Finally, it would be very useful to develop a procedure to evaluate the PI over an urban area where some monitoring stations are present and measure concentration of different pollutants. This is not a simple task. Concentration levels of the same pollutant, measured in an urban area, can differ strongly among the different stations since they are localised in different type of sites.

Table 7
Application of PI, proposed in this paper, to data measured in Naples assuming additive effect of pollutants

Pollution category	2001	2002
Good quality	4.3	0.0
Low pollution	33.8	45.2
Moderate pollution	50.6	44.7
Unhealthy for sensitive groups	1.4	9.1
Unhealthy	9.8	2.2

Percentage of occurrence of pollution categories.

Generally on the basis of their location, the monitoring stations in an urban area can be schematically defined as: (i) background stations; (ii) residential-area stations; (iii) high traffic road stations and (iv) photochemical stations.

Two different pollution indexes on an urban area could be defined: PI of a single pollutant p over the urban area (UPI_p) and PI of all pollutants over the urban area (UPI).

In both cases the simplest way to evaluate the PI of the urban area would be to assume the maximum or the average value of pollution indexes measured by the different stations. Both procedures would give misleading results. In the first case an excessive high PI would be attributed. In the second case the average could hide the presence of zones in the urban area characterised by high pollution levels and/or the fact that limit values established can be exceeded. A more rigorous procedure is necessary. If we assume that each station is representative of a situation of air pollution in the urban area, then a weight could be attributed to each station corresponding to the percentage of the urban area surface represented by that station. Moreover, it must be considered that in many cases the stations belonging to the same monitoring network do not include the same sensors: they do not measure the same pollutants. Therefore, a weight could be defined for each sensor with reference to pollutant p:

$$W_{s,p} = \frac{\text{Area represented by sensor measuring } p \text{ in station "}s\text{"}}{\text{Total urban area}} \tag{5}$$

and

$$\sum_{s=1}^{k} W_{s,p} = 1 \quad \forall\, p, \tag{6}$$

where k is the number of monitoring stations were pollutant p is measured. The calculation in Eq. (6) must be carried out considering only stations measuring

pollutant p. The urban PI of the pollutant p could be evaluated by:

$$\mathrm{UPI}_p = \sum_{s=1}^{k} \mathrm{PI}_{s,p} W_{s,p}. \tag{7}$$

In a similar manner a weight for each station could be defined independent of the pollutant:

$$W_s = \frac{\text{area represented by station } s}{\text{total urban area}} \quad \text{and} \quad \sum_{s=1}^{m} W_s = 1 \tag{8}$$

where m is the number of all the stations belonging to the network. Then the urban area pollution index UPI could be evaluated as:

$$\mathrm{UPI} = \sum_{s=1}^{m} \mathrm{PI}_s W_s. \tag{9}$$

It must be observed that if all the stations do not measure the same pollutants (i.e.; $m \neq k$), and this is the typical case, usually $W_s \neq W_{s,p}$.

For the urban area, even though the results obtained using urban pollution indexes defined in Eqs. (7) and (9) are more reliable than those obtained assuming the maximum or the average values of pollution indexes of single stations or sensors, two problems still remain in their application: (i) to attribute exact values to weights W is not straightforward; (ii) even though limit values in one or more stations would be exceeded the use of Eq. (7) or (9) could give a UPI_p or a UPI value corresponding to "good" or "low pollution".

The attribution of exact values to the weight W of each station or sensor requires a good knowledge of the urban area and the use of other information such as traffic flows, emission inventories, geometrical dimension of the roads and so on. The second problem could be solved imposing after the evaluation of UPI the condition $\mathrm{UPI} > 50$ if LV is exceeded in one or more stations.

4. Conclusions

The pollution index proposed seems adequate to measure air quality in the urban area of Naples. Local considerations cannot be ignored in applying air quality or pollution indexes. This is the reason why it has been demonstrated that EPA AQI is not appropriate for a European urban area like Naples. The need for a synthesis of pollutant concentration data collected by air monitoring networks calls for a development of air quality or pollution indexes. The procedure adopted and the results reported could make a contribution in this field. The higher or lower validity of a pollution index with respect to another can be proved only by comparing results of epidemiological studies with patterns of the pollution indexes for a significant range of time. Finally, if a pollution index is evaluated for single pollutant, the air quality probably is not exactly evaluated and the effect on human health could be underestimated.

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